Global to Local: A Hierarchical Detection Algorithm for Hyperspectral Image Target Detection

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***Abstract*— Hyperspectral image (HSI) has received considerable attention in the field of target detection due to its powerful ability to capture the spectral information of land covers, and plenty of detection algorithms have been explored. However, these methods generally leverage the difference between the spectrum of the target to be detected and the background spectrum to accomplish target detection, and so are susceptible to the problem of spectral variability. In this article, we propose a global to local hierarchical detection algorithm for HSI (G2LHTD). First, extended morphological attribute profile (EMAP) is first used to model global spatial texture information from HSI. Subsequently, a diverse-direction constrained energy minimization (D^2CEM) detector is developed to consider the spatial information within eight neighborhoods around each pixel in HSI, yielding comprehensive local spatial information. More substantially, to effectively discriminate the neighborhood information in diverse directions, we devise an adaptive neighborhood feature aggregation (ANFA) strategy, which will comprehensively evaluate the significance of neighborhood information in diverse directions. As a result, the spatial features of HSI can be comprehensively considered for hyperspectral target detection (HTD). Extensive experiments, conducted on four standard datasets, demonstrate the effectiveness of the proposed method. The codes of this work will be available at https://github.com/zhonghaochen/G2LHTD_Master for the sake of reproducibility.**

***Index Terms*— Adaptive neighborhood feature aggregation (ANFA), diverse-direction constrained energy minimization (D^2CEM), extended morphological attribute profile (EMAP), global to local, hyperspectral image (HSI), target detection.**

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I. Introduction

HYPERSPECTRAL image (HSI) has been widely studied in the field of Earth observation (EO) due to its powerful description of the spatial distribution of land covers along with their rich spectral information. Particularly, the spectral absorption variability induced by the physical and chemical properties of different land covers presents an opportunity for precise identification of land covers typologies [1], [2], [3], [4], [5]. On this basis, the target of interest detection algorithms based on HSI have become one of the vibrant research topics in HSI applications [6]. As opposed to target detection based on high-resolution remote sensing images that simply use the spatial structure features of land covers, the abundant spectral information possessed by HSI has unique advantages in the identification of small targets, camouflaged targets, and targets in complex backgrounds [7], [8], [9]. In recent years, HSI target detection (HTD) has been broadly applied in the fields of mineral detection, lesion diagnosis, military reconnaissance, and environmental monitoring [10], [11], [12].

Fundamentally, HTD tasks can be viewed as a binary classification problem, where the goal is to classify all pixels in HSI into the target category and the background category. However, dissimilar to the identification (classification) of all land covers of interest, the detected target in the HTD tasks often belongs to only a minor part of the imaging area, and the pixels of the other categories are all referred to as "background" class. It can be noticed that the target category exists in a sparse form [13]. Therefore, how suppressing the complex and diverse backgrounds classes from HSI to highlight the detected target is a key in HTD.

In the early phase of research on HTD algorithms, the difference between the spectral curve of the target and other land covers was adopted directly as a basis for detection [14]. This discrepancy is generally obtained by employing the calculation of the Euclidean distance (ED) [7], spectral information divergence (SID) [15], and spectral angle mapping (SAM) [16] of the detected pixel to the priori target spectrum. Obviously, this type of detection is simple and generic. However, due to the prevalence of subpixel targets caused by the low spatial resolution of HSI, it is difficult to obtain satisfactory performance by directly acquiring the spectral discrepancy [6], [17]. In response, a number of statistical-based methods had been further proposed. For example, the adaptive coherence

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estimator (ACE) [18] and the matched filter (MF) [19] allow us to understand the target detection task as a binary hypothesis testing problem based on a Gaussian distribution. Subsequently, some variants of MF were gradually developed, such as adaptive matched filter (AMF) [20] and spectral matched filter (SMF) [21]. Furthermore, methods based on constrained energy minimization (CEM) [22] are further investigated for the purpose of highlighting targets and suppressing background information. Xue et al. [23] designed the iterative CEM by applying the CEM to a Gaussian-filtered feature map, yielding high detection performance in the final detection. Moreover, other CEM-based detection algorithms have been further developed to boost the detection accuracy, including hierarchical CEM (hCEM) [24] and ensemble-based CEM (ECEM) [25]. Although the effectiveness of the aforementioned work in HTD has been demonstrated, the detection performance still remains limited due to the lack of representational power for spatial information, especially in scenarios with a poor spectral discrepancy between target and background.

As mentioned above, the detected target exists in a sparse form in HSI, and therefore machine learning-based methods [26] represented by sparse representation target detectors (STDs) are widely explored. For instance, Chen et al. [27] achieved satisfactory detection results by constructing a complete dictionary of target and background pixels and then figuring out the sparse vector. Based on this, some STD-based derived detectors have been gradually investigated. Wei et al. [28] constructed a combined sparse and collaborative representation (CSCR) for HTD, yielding high detection performance. Zhang et al. [29] proposed a detection algorithm based on sparse representation of binary hypothesis (SRBBH), which employs different sparse representations of pixels under two different hypotheses, thus boosting the target-background discrepancy. However, these methods still do not exploit the spatial information in HSI, due to the fact that the spatial association information between pixels was seldom considered [30].

In recent years, deep learning (DL)-based models have started to be tried in the field of HSI processing due to their excellent feature description capability [31], [32], [33], [34], [35]. However, the scarce target samples can hardly meet the demand for parameter training in DL-based HTD models. Therefore, the primary problem facing the current DL-based HTD algorithm is how to augment the training samples. For example, Du et al. [36] achieved the augmentation of training samples by performing linear operations between pixels and thus implemented HTD using a deep convolutional neural network (CNN). Similarly, building pixel pairs by combining target and background pixels has also been shown to enable training sample augmentation [37], [38]. Besides augmenting target category samples, the generative adversarial network (GAN) [39], [40], is also utilized to learn the background distribution to highlight the targets of interest. More significantly, taking spatial features into account for HTD is increasingly becoming a trend in current research on detection algorithms [41]. For instance, Feng et al. [42] post-processed the detection results to extract spatial context information, yielding excellent performance. Recently, Xie et al. [43] constructed a structure-to-structure selection rule to fully consider the spatial features of HSI, and finally fused the extracted spatial-spectral features at the decision level for detection. Moreover, Qin et al. applied the vision transformer (ViT) [44] to learn the spectral-spatial features of each pixel and its neighboring pixels, which can effectively discover the potential context information in HSI. Nevertheless, these detectors only treat images as ensembles of spectral without taking advantage of their spatial structure, and their detection results are predominantly contributed by spectral information. More recently, there are some algorithms that take local spatial structural features into account. For example, Yang et al. [45] extracted spatial and spectral information simultaneously by patch-wise learning with 3D-CNN. Cao et al. [46] developed a novel autoencoder by employing the 2-D convolutional kernel, which can better utilize local spatial features to yield better background construction. By further considering the diversity of the background structure, Cao et al. [47] obtained a more realistic background distribution by reconstructing the background using dilated convolution. Although the above methods provide promising detection accuracy using spatial features, the use of only local spatial information may lead to local background false alarms. Moreover, DL-based methods typically suffer from high-computational complexity, which is unfavorable to the in-orbit processing of data [48].

Extended morphological attribute profile (EMAP) [49] is considered as a morphological feature acquisition algorithm, which can efficiently obtain the structural information of elements in an image through multiple morphological attribute filters (AFs) [50]. Consequently, EMAP can be naturally employed to model the global spatial information of HSI. In the field of HTD, there are some related attempts to utilize EMAP. Imani [51] proposed to combine EMAP with collaborative representation and sparse representation for HTD, respectively. Furthermore, Wu et al. [52] incorporated EMAP with low-rank multitask learning, obtaining effective background suppression. More recently, Huang et al. [53] constructed a background purification framework by employing EMAP to highlight targets better. However, there are some potential challenges of hyperspectral target detectors in terms of spatial feature extraction.

1) The low spatial resolution of the HSI is a significant bottleneck limiting the performance of the detectors, especially when the detected targets are small. Effective use of fine spatial structure information becomes particularly necessary.
2) EMAP is able to sketch all texture information in the whole image, that is, all structure information is described. This might cause background information to be accentuated, which is negative to target recognition.
3) Last but not least, the utilization of spatial information only suppresses the background from a global perspective thus neglecting the detection of the local edge information of the target, which has a major influence on the detection performance.

To handle these challenges, in this work, a global to local hierarchical detection algorithm is proposed for HTD

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(called G2LHTD). Different from existing methods, G2LHTD extracts different levels of spatial features in HSIs hierarchically, thus better highlighting the target and suppressing the background. The proposed G2LHTD algorithm consists of three main parts. First, EMAP is adopted to extract multilevel and multiattribute spatial features to obtain an EMAP dataset with global spatial information. Subsequently, to fully utilize the local spatial contextual information around the pixel and alleviate the EMAP causing some background information to be highlighted, we developed a diverse-direction constrained energy minimization detector (called D^2CEM) based on the conventional CEM detector by considering the information of the eight neighborhoods around the test pixel. Furthermore, although it is known that neighboring pixels tend to belong to the same feature class, the low spatial resolution of the HSI might pose the problem of the large variability in the neighborhood information. This naturally drives us to adaptively utilize neighborhood information in different directions by developing a novel fusion strategy, making them even more appropriate for HTD. More specifically, the major contributions of this article can be unfolded as follows.

1) A novel spatial information investigation algorithm, called G2LHTD, is developed for HTD by hierarchically exploring the discrepancy of target and background.
2) The EMAP is adopted to model global spatial feature of HSI. It allows a flexible investigation of HSI and a more complete description of the scene can be obtained. Accordingly, the background and the target will be separated to different degrees, which will facilitate further exploration of local information subsequently.
3) To explore the local neighborhood information of the test pixel more comprehensively and mitigate the interference of background information, D^2CEM detector is proposed to further accurately evaluate the texture information in profiles.
4) To further improve the discriminative power of the contribution of neighborhood information in the detection results, a newly adaptive neighborhood feature aggregation (ANFA) strategy is proposed, in which the relationship between the neighborhood pixels, test pixel, and a prior target pixel is well exploited.

The rest parts of this article are organized as follows. Section II briefly reviews EMAP-related knowledge. Section III presents the workflow of the proposed G2LHTD and then details every step of it. In Section IV, extensive experiments are implemented on four real datasets, and the results are discussed qualitatively and quantitatively. Finally, we summarize the work of this article in Section V.

II. Related Work

In this section, some preliminaries of EMAP are provided by reviewing the fundamental definitions, including morphological profile (MP) and attribute profile (AP).

A. Morphological Profile (MP)

There are rich literatures indicate that spatial texture features of HSI play a vital role in boosting the recognition accuracy of objects of interest [54], [55], [56], [57]. Particularly, due to the diversity and complexity of the target structures, MP is proposed to explore the structural response under different scale hypothesis spaces. Specifically, MP enables the highlighting of different spatial texture information of the land covers by repeatedly using structuring elements (SEs) of fixed shape and increasing size on the image [50]. First, the erosion and dilation of an image $\boldsymbol{X}$ by a SE $\boldsymbol{B}$ is denoted by $\varepsilon_B(\boldsymbol{X})$ and $\delta_B(\boldsymbol{X})$, respectively

$$\varepsilon_B(\boldsymbol{X}) = \{x \mid \boldsymbol{B}_x \subseteq \boldsymbol{X}\} \tag{1}$$

$$\delta_B(\boldsymbol{X}) = \{x \mid \boldsymbol{B}_x \cap \boldsymbol{X} \neq \varnothing\}. \tag{2}$$

As shown in formula (1), the erosion of an image $\boldsymbol{X}$ is defined as the locus of points, x, such that $\boldsymbol{B}$ is included in $\boldsymbol{X}$ when its origin is placed at x. Similarly, as shown in formula (2), the dilation of an image $\boldsymbol{X}$ is defined as the locus of points, x, such that $\boldsymbol{B}$ hits $\boldsymbol{X}$ when its origin coincides with x [58].

Then, on the basis of the image erosion and dilation, the definitions of opening by reconstruction γ_R and closing by reconstruction ϕ_R, of a grayscale image $\boldsymbol{f}$, are formulated by

$$\gamma_R^i(\boldsymbol{f}) = R_f^{\delta}\left[\varepsilon^i(\boldsymbol{f})\right] \tag{3}$$

$$\phi_R^i(\boldsymbol{f}) = R_f^{\varepsilon}\left[\delta^i(\boldsymbol{f})\right]. \tag{4}$$

An opening by reconstruction γ_R is defined as the erosion of the grayscale image $\boldsymbol{f}$ with a SE of size i, followed by a reconstruction by dilation. By duality, a closing by reconstruction ϕ_R, is defined as the dilation of the grayscale image $\boldsymbol{f}$ with a SE of size i, followed by a reconstruction by erosion [58].

Finally, a MP is a concatenation of morphological closing profiles and morphological opening profiles, which are a sequence of closings and openings by reconstruction with a SE of fixed shape and increasing size $\boldsymbol{i} \in [0, \ldots, n]$, respectively

$$\mathrm{MP}(\boldsymbol{f}) = \left\{ \underbrace{\phi_R^n(\boldsymbol{f}), \phi_R^{n-1}(\boldsymbol{f}), \ldots, \phi_R^1(\boldsymbol{f})}_{\text{morphological closing profiles}}, \boldsymbol{f} \times \underbrace{\gamma_R^1(\boldsymbol{f}), \ldots, \gamma_R^{n-1}(\boldsymbol{f}), \gamma_R^n(\boldsymbol{f})}_{\text{morphological opening profiles}} \right\}. \tag{5}$$

After the MP operation, the texture features such as gradient, minima, and maxima regions in the image can have been highlighted, which brings great convenience to the subsequent feature extraction. However, the MP operation has the following limitations. First, the shape of SEs is fixed; second, SEs cannot describe information related to regional grayscale features, such as spectral uniformity and contrast; third, the computational complexity is high [59].

B. Attribute Profile (AP)

To tackle the above issues, AP is proposed. Unlike MP that use SEs, AP performs operations on connected components (CCs) through AFs, where CCs do not have a specific shape, so AP can overcome the drawback of fixed shape of SEs. Meanwhile, AP is able to set CCs to the grayscale values of

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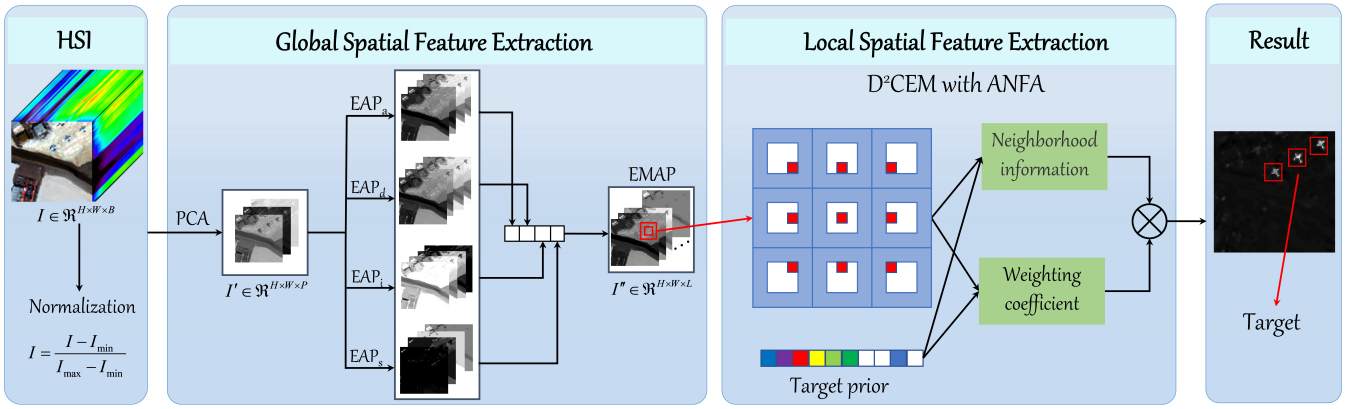

Fig. 1. Flowchart of the proposed G2LHTD. First, the original HSI data is normalized. Then, EMAP is used to model global spatial features to obtain EMAP datasets with global spatial information. Next, in the local spatial feature extraction stage, the neighborhood information is obtained by D^{2}CEM detector, and the ANFA strategy is proposed to obtain the weighting coefficient. Finally, the neighborhood information is weighted to obtain the final detection.

adjacent regions by comparing the magnitude of their attributes with the filter reference values, so AP is enabled to describe information related to the grayscale characteristics of regions. Additionally, AP is more computationally efficient than MP because it is based on the Max-tree representation [59].

Different from MP, which is based on morphological opening and closing profiles, AP is based on attribute opening and closing profiles. For a given grayscale image $\boldsymbol{f}$, attribute opening profiles and attribute closing profiles can be represented as

$$\prod_{\phi^{T_\lambda}}(\boldsymbol{f}) = \phi^{T_\lambda}(\boldsymbol{f}) \quad \forall\lambda \in [0,..,l] \tag{6}$$

$$\prod_{\gamma^{T_\lambda}}(\boldsymbol{f}) = \gamma^{T_\lambda}(\boldsymbol{f}) \quad \forall\lambda \in [0,..,l] \tag{7}$$

where ϕ_λ^T and γ_λ^T, respectively, represent the morphological attribute closing operation and morphological attribute opening operation on the CCs in a grayscale image $\boldsymbol{f}$, with a series of increasing criteria $\boldsymbol{T} = \{T_\lambda : \lambda = 0, 1, \ldots, l\}$ [59].

Similar to MP, AP is obtained by concatenating the attribute opening profiles and attribute closing profiles in the following form:

$$AP(\boldsymbol{f}) = \Big\{ \underbrace{\phi^{T_{\lambda_l}}(\boldsymbol{f}), \phi^{T_{\lambda_{l-1}}}(\boldsymbol{f}), \ldots, \phi^{T_{\lambda_1}}(\boldsymbol{f})}_{\text{attribute closing profiles}}, \boldsymbol{f} \times \underbrace{\gamma^{T_{\lambda_1}}(\boldsymbol{f}), \ldots, \gamma^{T_{\lambda_{l-1}}}(\boldsymbol{f}), \gamma^{T_{\lambda_l}}(\boldsymbol{f})}_{\text{attribute opening profiles}} \Big\}. \tag{8}$$

III. Proposed Method

In this section, the specific parts of the proposed G2LHTD will be discussed in detail. The flowchart of the G2LHTD is shown in Fig. 1. In Section III-A, we first explain what role of EMAP plays in the proposed method. Then, in Section III-B, the proposed local spatial feature extractor called D^{2}CEM, which is based on dual window, will be described. Finally, in Section III-C, the ANFA is introduced, yielding a general local neighborhood spatial feature fusion algorithm. The specific implementation of the G2LHTD is described in algorithm 1.

Algorithm 1 G2LHTD

Inputs: The original HSI $\boldsymbol{I} \in \mathbb{R}^{H\times W\times B}$.

Output: The detection result of the proposed method $\boldsymbol{Y}$.

1: Normalize the original HSI data via maximum and minimum normalization.
2: Down-dimension the normalized HSI data by PCA to obtain $\boldsymbol{I'} \in \mathbb{R}^{H\times W\times P}$.
3: Apply AP on each principal component using four different attributes (area of the region, length of the diagonal of the bounding box, moment of inertia, and standard deviation) respectively, and stack to form EMAP dataset $\boldsymbol{I''} \in \mathbb{R}^{H\times W\times L}$.
4: Send EMAP dataset and the prior target to D^{2}CEM detector, and dual window CEM dectection results of the test pixel and its eight neighboring pixels $\hat{\boldsymbol{Y}}_9 = [y_{(i,j)}, y_{(i-1,j)}, y_{(i+1,j)}, \ldots, y_{(i+1,j-1)}]$ are obtained.
5: Calculate the weighting coefficients of the test pixel and its eight neighboring pixels $\boldsymbol{W}_{SAP} = [(\rho_{\boldsymbol{p}_{(i,j)}\boldsymbol{d}}/s_{\boldsymbol{p}_{(i,j)}\boldsymbol{d}}), \rho_{\boldsymbol{p}_{(i-1,j)}\boldsymbol{p}_{(i,j)}}, \rho_{\boldsymbol{p}_{(i+1,j)}\boldsymbol{p}_{(i,j)}}, \ldots, \rho_{\boldsymbol{p}_{(i+1,j-1)}\boldsymbol{p}_{(i,j)}}]$ through Pearson correlation coefficient and SAM.
6: Obtain the final detection result $\boldsymbol{Y} = \boldsymbol{W}_{SAP} \times \hat{\boldsymbol{Y}}_9^T$ by multiplying the results of 4 and 5 correspondingly.

A. Global Spatial Features Obtain

As discussed previously, the key to HTD is to effectively suppress the background and highlight the target. However, compared to the singleness of the target class, the background class often contains a large amount of complex heterogeneous information, which has been a thorny problem-facing target detection research. Inspired by [52], in this section, we discuss the use of EMAP to model global spatial features of HSI to achieve the preliminary separation of background and target from a spatial perspective. As mentioned in the related work, AP can evaluate all CCs in an image to achieve merging of similar components. Inspired by this, we try to investigate whether this can reduce the complexity of background texture features by merging background structures with similar

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grayscale to some extent. Simultaneously, the target will be accentuated in different profiles thus achieving a preliminary separation from the background due to the morphological difference with the background (see Fig. 2). Accordingly, AP is naturally considered for modeling the global spatial structure information of HSI. However, since AP can only be used for grayscale images, the extended structure of AP, extended attribute profile (EAP), was proposed to be applied to HSIs with high-dimensional properties [59]. As shown in Fig. 1, the input original HSI $\boldsymbol{I} \in \mathbb{R}^{H\times W\times B}$ is first down-dimensioned by principal component analysis (PCA) to obtain $\boldsymbol{I'} \in \mathbb{R}^{H\times W\times P}$, which contains the first P principal components (PCs) of the spectral features. Subsequently, the AP of each PC can be calculated and then the EAP can be simply stacked as

$$\mathrm{EAP}\left(\boldsymbol{I'}\right) = \left\{AP(\boldsymbol{PC_1}), AP(\boldsymbol{PC_2}), \ldots, AP(\boldsymbol{PC_n})\right\}. \quad (9)$$

Due to the diversity of the spatial structure of land covers, this necessitates the use of different criteria to estimate the CC. Therefore, EAP with different attributes were used to model the spatial information of the PCs. Subsequently, the results after being modeled by different attribute EAP are stacked to obtain the following EMAP with multiattribute spatial characterization:

$$\mathrm{EMAP}\left(\boldsymbol{I'}\right) = \left\{\mathrm{EAP}_{a_1}\left(\boldsymbol{I'}\right), \mathrm{EAP}_{a_2}\left(\boldsymbol{I'}\right), \ldots, \mathrm{EAP}_{a_k}\left(\boldsymbol{I'}\right)\right\} \quad (10)$$

where k denotes the results under k different attributes were obtained. The attributes commonly used in the field of HSI processing are area of the region, length of the diagonal of the bounding box, moment of inertia, and standard deviation [59]. Consequently, the EMAP with a global spatial information description is obtained (see Fig. 2), denoted as $\boldsymbol{I''} \in \mathbb{R}^{H\times W\times L}$, where L represents number of profiles.

By investigating the whole scene flexibly as described above, background and target information at different levels can be obtained elegantly. Moreover, after this process, more refined detection can be effectively implemented subsequently.

B. Local Spatial Features Obtain

As is known to all, reducing mutual target and background interference is crucial in HTD. Accordingly, the dual window target detector [60] shown in Fig. 3 is commonly used as a basic detection strategy. Specifically, the local spatial features are excavated using two concentric square windows of different sizes centered on the test pixels. Thus, pixels with high probability of being targets in the inner window are protected while the area between the inner and outer window could be considered as pure background pixels can be obtained. Furthermore, these local background pixels will be employed to generate the purer background autocorrelation matrix.

However, due to the diversity of the target spatial structure, in addition to obtaining the information contained in the dual window centered on the test pixel, its neighborhood information should also be fully considered. In this section, therefore, the D^{2}CEM detector based on the original dual

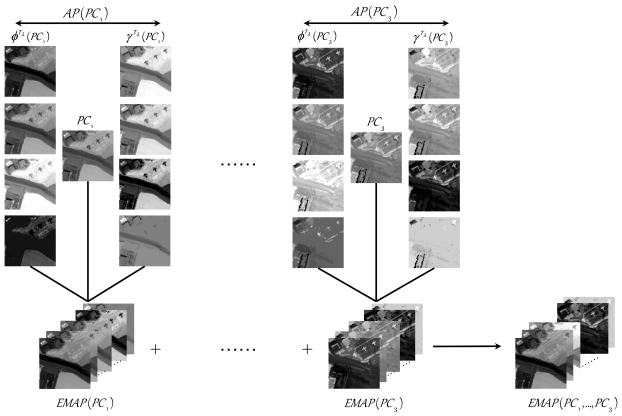

Fig. 2. General architecture of EMAP. Calculate AP on first three PC s using different attributes, respectively, and stack to form EMAP dataset.

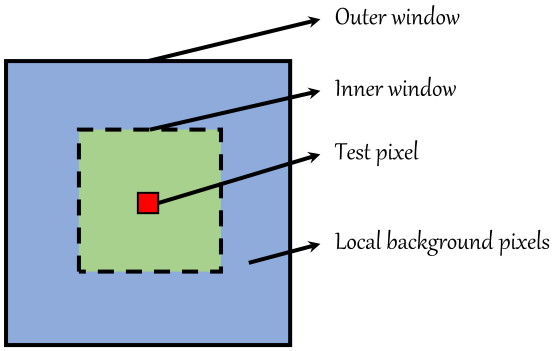

Fig. 3. Dual window target detection strategy.

window detector is proposed. In this way, the dual window information of the eight neighboring pixels of the test pixel is considered to compensate for the inadequate use of local spatial information brought about using only the central pixel, especially when facing the detection of complex targets.

We first define $\boldsymbol{p}_{(i,j)}$ to represent the pixel being tested of $\boldsymbol{I''}$, then its eight neighboring pixels can be accordingly denoted as $\boldsymbol{p}_{(i-1,j)}$, $\boldsymbol{p}_{(i+1,j)}$, $\boldsymbol{p}_{(i,j+1)}$, $\boldsymbol{p}_{(i,j-1)}$, $\boldsymbol{p}_{(i-1,j+1)}$, $\boldsymbol{p}_{(i+1,j+1)}$, $\boldsymbol{p}_{(i-1,j-1)}$, and $\boldsymbol{p}_{(i+1,j-1)}$, where i and j represent the horizontal and vertical coordinates, respectively. When a dual window detector with inner and outer windows of size win_{in} and win_{out} is applied to the test pixel $\boldsymbol{p}_{(i,j)}$, the pixels between the inner and outer windows are approximated as background pixels, which can be denoted as

$$\boldsymbol{X}_n^{(i,j)} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n] \quad (11)$$

where n represent the number of local background pixels, which can be calculated as

$$n = \mathrm{win}_{\mathrm{out}} \times \mathrm{win}_{\mathrm{out}} - \mathrm{win}_{\mathrm{in}} \times \mathrm{win}_{\mathrm{in}}. \quad (12)$$

Subsequently, these approximate background pixels will be combined with a prior target pixel $\boldsymbol{d}$ to construct a CEM filter, which will be adopted to detect the center pixel $\boldsymbol{p}_{(i,j)}$. More specifically, the autocorrelation matrix $\boldsymbol{R}_n^{(i,j)}$ of $\boldsymbol{X}_n^{(i,j)}$ can be

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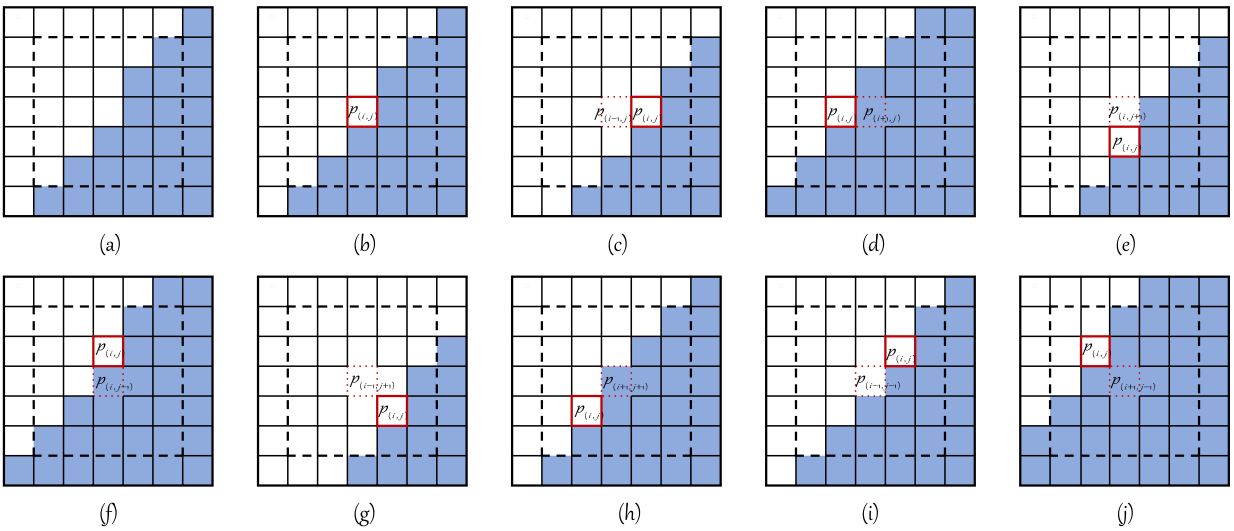

Fig. 4. Illustration of the dual window of central test pixel and its eight neighboring pixels. The white grid and the blue grid represent background pixel and target pixel, respectively. The red solid line box represents the test pixel, while the red dotted line box represents the neighboring pixel. (a) Dual window. (b) Central. (c) Left-direction. (d) Right-direction. (e) Up-direction. (f)Bottom-direction. (g) Upleft-direction. (h) Upright-direction. (i) Bottomleft-direction. (j) Bottomright-direction.

formulated as

$$\boldsymbol{R}_n^{(i,j)} = \frac{1}{n}\sum_{m=1}^{n} \boldsymbol{x}_m \boldsymbol{x}_m^T. \tag{13}$$

Then, the filter vector of CEM can be expressed as

$$\boldsymbol{w} = \frac{\boldsymbol{R}_n^{-1}\boldsymbol{d}}{\boldsymbol{d}^T \boldsymbol{R}_n^{-1}\boldsymbol{d}} \tag{14}$$

where $\boldsymbol{w}$ need to meet $\boldsymbol{w}^T\boldsymbol{d} = 1$, which is the construction condition of CEM filter here. Consequently, the detection result of $\boldsymbol{p}_{(i,j)}$ can be obtained as

$$y_{(i,j)} = \boldsymbol{w}^T \boldsymbol{p}_{(i,j)}. \tag{15}$$

Then, the dual window detector is moved around the test pixel until all eight neighboring pixels of the pixel being detected have been detected. Consequently, eight detection results of eight directions were obtained, that is $y_{(i-1,j)}$, $y_{(i+1,j)}$, $y_{(i,j+1)}$, $y_{(i,j-1)}$, $y_{(i-1,j+1)}$, $y_{(i+1,j+1)}$, $y_{(i-1,j-1)}$, and $y_{(i+1,j-1)}$. As a result, more comprehensive local neighborhood detection results can be obtained as follows:

$$\hat{\boldsymbol{Y}}_9 = \left[y_{(i,j)}, y_{(i-1,j)}, y_{(i+1,j)}, \ldots, y_{(i+1,j-1)}\right]. \tag{16}$$

Illustratively, the procedure of D^{2}CEM is shown in Fig. 4. From Fig. 4, we can observe that the advantages of the proposed D^{2}CEM predominantly lie in two aspects: First, the proposed D^{2}CEM can provide a wider neighborhood view. Second, spatial correlations in different directions can be better explored later with such a larger scope. It can be found that the dual window region where the test pixel $\boldsymbol{p}_{(i,j)}$ is located is composed of an irregular target and background region, which is a very common phenomenon when detecting the boundary pixels of target. If we only use the dual window detector centered on the pixel to be tested, too much heterogeneous information may be introduced, which will lead to wrong judgment. Conversely, when using the diverse-direction detection strategy, there is at least one direction can contain purer information, which could facilitate the judgment of the pixel being tested to some extent.

C. Local Spatial Feature Adaptive Fusion Strategy: ANFA

Although adjacent pixels are approximated to be in the same category due to their geographical proximity, the low spatial resolution of HSIs is susceptible to more uncertainty (e.g., mixed pixels). Consequently, neighborhood information in different directions may not provide the same contribution to the detection results of the pixel under test. As a result, it is natural to consider how to make reasonable use of the eight-neighborhood detection results obtained in Section III-B to improve the detection performance.

In this section, we newly propose a strategy (called ANFA) to adaptively fuse the detection results of the pixel under test obtained in Section III-B and the detection results of its individual neighbors. First of all, two spectral similarity measurement methods are introduced, respectively, which will be harnessed to evaluate the correlation between the test pixel, the neighborhood information, and a prior target pixel. The first one is the Pearson correlation coefficient, whose value is proportional to the similarity of the two spectral. In addition, the SAM measures the spectral similarity by calculating the angle between two spectral, and thus its value is inversely proportional to the similarity. They can mathematically express as follows (17) and (18), shown at the bottom of the next page, where $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are two different vectors, L is the number of vectors, $\rho_{\boldsymbol{x}_1\boldsymbol{x}_2}$ represents the Pearson correlation coefficient of the two vectors, and $s_{\boldsymbol{x}_1\boldsymbol{x}_2}$ is the SAM value of the two vectors.

Subsequently, a newly adaptive weight acquisition strategy, spectral angel, and Pearson joint weighting (SAP), is explored

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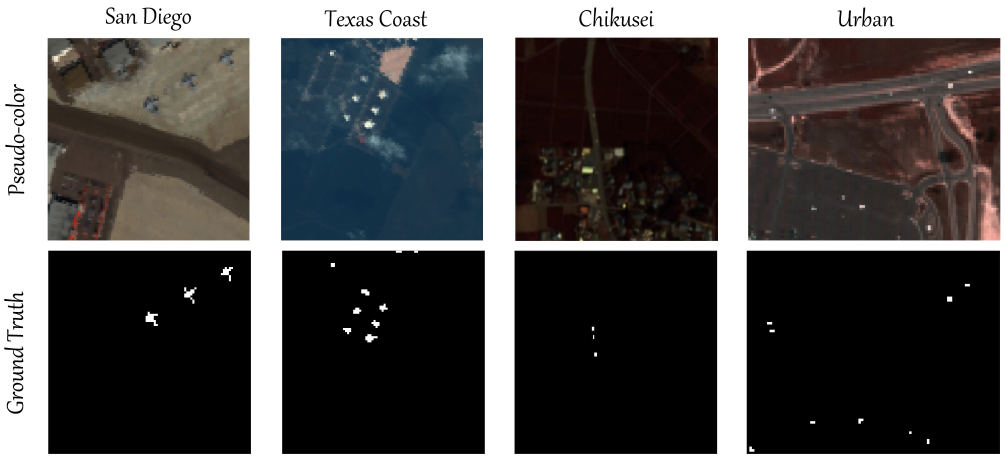

Fig. 5. Pseudo-color images and according groud truth maps of four considered HSI datasets.

to be able to elegantly exploit the neighborhood information. The weighting coefficients can be attained flexibly using Pearson coefficients and SAM, which can be mathematically expressed as follows:

$$\boldsymbol{W}_{\mathrm{SAP}} = \left[\frac{\rho_{\boldsymbol{p}_{(i,j)}\boldsymbol{d}}}{s_{\boldsymbol{p}_{(i,j)}\boldsymbol{d}}}, \rho_{\boldsymbol{p}_{(i-1,j)}\boldsymbol{p}_{(i,j)}}, \rho_{\boldsymbol{p}_{(i+1,j)}\boldsymbol{p}_{(i,j)}}, \ldots, \rho_{\boldsymbol{p}_{(i+1,j-1)}\boldsymbol{p}_{(i,j)}} \right] \tag{19}$$

where $\boldsymbol{p}_{(i,j)}$ and $\boldsymbol{d}$ denote the pixel being test and prior target pixel, respectively. $\boldsymbol{p}_{(i\pm1,j\pm1)}$ refers to eight neighboring pixels of test pixel. Different from the commonly used average similarity between neighboring pixels and a priori target pixel to determine the contribution of neighborhood information to the detection result of the test pixel, which does not sufficiently take into account the influence of each direction individually. Thus, we attempt to separately consider the effect of neighborhood information in different directions on the determination of the test pixel and aggregate neighborhood features adaptively. As constructed in (19), a weight acquisition method that is weighted according to the significance of the neighborhood in each direction is introduced. Specifically, since the variability of neighboring spectral information exists in HSI, the correlation between neighboring pixels in each direction and the test pixel is measured by the Pearson coefficient as a weight measure of the importance of neighboring information in that direction. Furthermore, the correlation between the test pixel and the prior target pixel should also be paid more attention. Accordingly, the correlation between the pixel under test and the prior target pixel is calculated, including both Pearson correlation coefficient and SAM. Then, the correlation between the pixel being tested and the prior target pixel can be calculated as $(\rho_{\boldsymbol{p}_{(i,j)}\boldsymbol{d}}/s_{\boldsymbol{p}_{(i,j)}\boldsymbol{d}})$. It can be observed that the test pixel will be particularly emphasized when the more similar it is to the prior target pixel, due to the opposite numerical trend of Pearson coefficient and SAM in characterizing the similarity. Moreover, the test pixel will be given a higher weight than the neighborhood pixels. In this way, the neighborhood pixels and the importance of the test pixel can be measured more appropriately. Finally, the target detection result under the effect of this weight can be obtained and can be formulated as

$$\boldsymbol{Y} = \boldsymbol{W}_{\mathrm{SAP}} \times \hat{\boldsymbol{Y}}_9^T \tag{20}$$

where $\hat{\boldsymbol{Y}}_9^T$ represents the transpose of $\hat{\boldsymbol{Y}}_9$.

IV. Experimental Results and Analysis

In this section, four well-known public HSI benchmarks are described first. The evaluation indicators and implementation details are then introduced. Please note that these four datasets were captured by three different imaging platforms in different scenes, which brings a great challenge to the generalizability of the algorithms. Subsequently, the ablation analysis is carried out on our proposed method. Finally, eight state-of-the-art detection algorithms are introduced and extensive experiments are conducted to evaluate the detection performance of proposed G2LHTD both quantitatively and qualitatively. All experiments are carried out with Python 3.7.11, Pytorch 1.10.0, and MATLAB R2016b on a laptop with an Intel Core i7-11800H CPU and NVIDIA GEFORCE RTX 3060 GPU.

$$\rho_{\boldsymbol{x}_1\boldsymbol{x}_2} = \frac{\sum \boldsymbol{x}_1{}^T\boldsymbol{x}_2 - \frac{(\sum \boldsymbol{x}_1)^T \sum \boldsymbol{x}_2}{L}}{\sqrt{\left(\sum \boldsymbol{x}_1{}^T\boldsymbol{x}_1 - \frac{(\sum \boldsymbol{x}_1)^T \sum \boldsymbol{x}_1}{L}\right)\left(\sum \boldsymbol{x}_2{}^T\boldsymbol{x}_2 - \frac{(\sum \boldsymbol{x}_2)^T \sum \boldsymbol{x}_2}{L}\right)}} \tag{17}$$

$$s_{\boldsymbol{x}_1\boldsymbol{x}_2} = \cos^{-1}\left(\frac{\boldsymbol{x}_1{}^T\boldsymbol{x}_2}{\sqrt{\boldsymbol{x}_2{}^T\boldsymbol{x}_2}\sqrt{\boldsymbol{x}_1{}^T\boldsymbol{x}_1}}\right) \tag{18}$$

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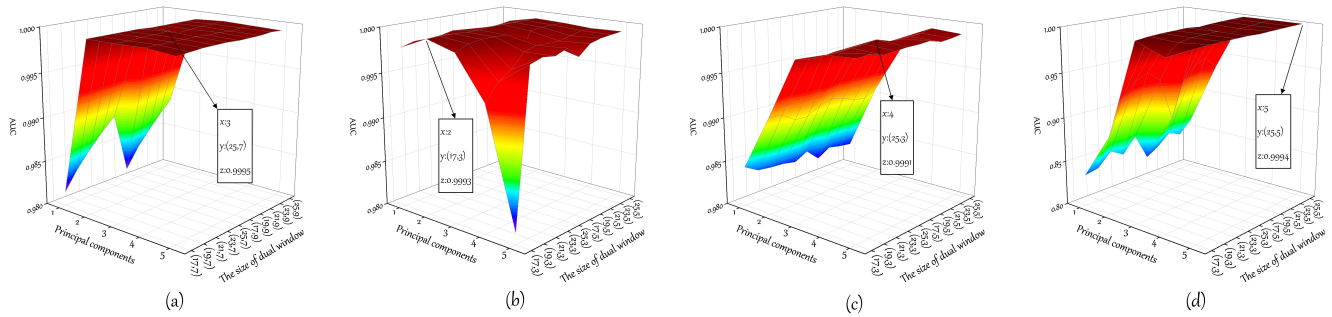

Fig. 6. Effect of P and $(\text{win}_{\text{out}}, \text{win}_{\text{in}})$ on four experimental HSI datasets. (a) San Diego. (b) Texas Coast. (c) Chikusei. (d) Urban.

A. Dataset Description

The pseudo-color images and ground truth (GT) maps of the four HSI considered datasets are shown in Fig. 5.

1) San Diego Dataset: The first HSI data were collected by airborne visible/infrared imaged spectrometer (AVIRIS) [61] over airport of San Diego, CA, USA. The HSI dataset covers 100 × 100 pixels at a ground sampling distance (GSD) of 3.5 m. After removing several water absorption attenuation bands, the other 189 high signal-to-noise ratio (SNR) bands are retained [62]. There are 57 pixels of three aircrafts for detected and several central pixels of them are adopted to prior target pixels.

2) Texas Coast Dataset: The second well-known benchmark was also acquired by AVIRIS over the coast area of Texas, USA, in 2010. The HSI cube comprises 100 × 100 pixels with GSD of 17.2 m and 204 high SNR channels [63], which has a relatively simple background to San Diego but lower GSD. The targets being detected in this scene are buildings, which are composed of 67 pixels.

3) Chikusei Dataset: The third benchmark HSI is captured in 2014 by Headwell HyperSpec-Vnir-C. This dataset describes a scene at the urban-rural junction of Chikusei City, Ibaraki Prefecture, Japan, where three targets of interest on a highway need to be detected, containing 15 pixels. The image consists of 100 × 100 pixels at GSD of 2.5 m with 128 high SNR spectral channels [64].

4) Urban Dataset: The fourth benchmark is Urban, which was collected using hyperspectral digital imagery collection experiment (HYDICE) sensor over the urban area of California, USA. The HYDICE sensor acquired 162 high SNR bands with 10 nm spectral resolution and 80 × 100 pixels with 2 GSD. The targets being detected in this scene are vehicles and buildings, which are composed of 21 pixels.

B. Assessment Criteria

To comprehensively analyze the performance of the detectors, three well-known indices are applied to evaluate the detection performance quantitatively.

1) Receiver Operating Characteristic (ROC): The receiver operating characteristic (ROC) curve is one of the widely used quantitative evaluation metrics in target detection, which describes the relationship between true positive rate (TPR) and false positive rate (FPR) in a right-angle coordinate system. Specifically, the x-axis and y-axis of the coordinate system record the FPR and the TPR, respectively, and thus the area on the right side of the curve represents the detection accuracy of the detector. Accordingly, for a promising detector, the ROC curve should be located as close as possible to the upper left corner of the coordinate system. The FPR and TPR can be directly formulated by

$$\text{FPR} = \frac{N_{FP}}{N_{FP} + N_{TN}}, \quad \text{TPR} = \frac{N_{TP}}{N_{TP} + N_{FN}} \tag{21}$$

where N_{FP} represents the number of background pixels detected as target pixels, N_{TP} indicates the number of target pixels detected, N_{FN} implies the number of target pixels marked as background pixels, and N_{TN} express the number of background pixels detected.

2) Area Under the ROC Curve (AUC): Since the ROC curves of various detectors are susceptible to overlap and their performance cannot be judged accurately enough, the area under the ROC curve (called AUC) is adopted to further quantitatively evaluate the performance of the detector [65]. The value of ROC is between 0 and 1, and the larger the AUC value, the better the performance of the corresponding detector.

3) Separability Map: Notoriously, the degree of separation between the target and the background is an essential indicator to demonstrate the effectiveness of the detection. In addition to the visual perception of how nicely the target is highlighted, quantitative analysis is equally requisite. The separability map [66] is a well-known statistical method used to depict the degree of separation, which utilizes two boxes to represent the range of normalized statistical values of the target and background values in the detection results, respectively. When the two boxes are farther apart in the y-axis direction, it indicates a better separation of target and background.

C. Analysis of Parameters

As mentioned before, to use EMAP to extract multilevel global spatial texture features, PCA is applied to reduce the spectral dimension of original HSI to obtain P principal components. In addition, the size of the two windows $(\text{win}_{\text{out}}, \text{win}_{\text{in}})$ in the proposed D^{2}CEM affects the degree of separation between the target and the background. Consequently, the selection of these two hyperparameters has a significant impact on the detection performance of the model.

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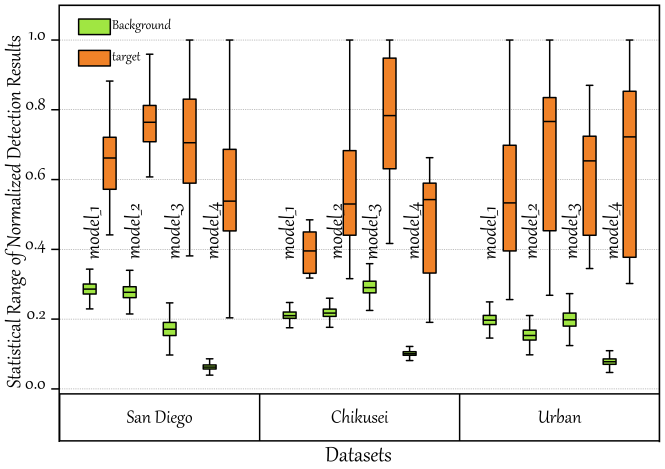

Fig. 7. Separability map of four models on the San Diego, Chikusei, and Urban datasets.

TABLE I

ABLATION ANALYSIS OF THE PROPOSED G2LHTD WITH A COMBINATION OF DIVERSE STRATEGIES ON THE SAN DIEGO, CHIKUSEI, AND URBAN DATASETS

Datasets	Model	EMAP	D^2CEM	Ave-Fusion	ANFA	AUC
San Diego	1	✗	✗	✗	✗	0.9930
	2	✓	✗	✗	✗	0.9974
	3	✓	✓	✓	✗	0.9986
	4	✓	✓	✗	✓	**0.9995**
Chikusei	1	✗	✗	✗	✗	0.9956
	2	✓	✗	✗	✗	0.9976
	3	✓	✓	✓	✗	0.9971
	4	✓	✓	✗	✓	**0.9991**
Urban	1	✗	✗	✗	✗	0.9963
	2	✓	✗	✗	✗	0.9975
	3	✓	✓	✓	✗	0.9955
	4	✓	✓	✗	✓	**0.9994**

In this scenario, it is indispensable to investigate proper range of these two parameters.

For the retained P principal components, we increment P from 1 to 5 and then analyze the effect of it on the detection results (AUC). Similarly, for the choice of outer and inner window sizes, the detection performance is explored for different combinations. The effects of the above two hyperparameters on the model detection performance simultaneously are reported in Fig. 6. It can be intuitively observed that the effect of number of PCs on detection results is similar for the four datasets. Specifically, when P is set too small, the spectral information cannot be retained sufficiently to obtain satisfactory detection results, and when too many PCs are retained, it leads to feature redundancy or even impairs the detection performance [see Fig. 6 (b)]. For win_{out} and win_{in}, as shown in Fig. 6, the proposed G2LHTD is not sensitive to the choice of size since the proposed D^2CEM and ANFA can fully excavate and flexibly utilize the neighborhood information. Considering the effect of two hyperparameters simultaneously on the proposed method performance, the P and $(\text{win}_{\text{out}}, \text{win}_{\text{in}})$ are set to be 3 and (25,7), 2 and (17,3), 4 and (25,3), and 5 and (25,5) for the San Diego, Texas Coast, Chikusei, and Urban dataset, respectively.

D. Ablation Study

We thoroughly investigate the detection performance gain of our proposed G2LHTD in terms detection accuracies using (in stepwise fashion) different strategies in models. Therefore, we carry out extensive ablation experiments on San Diego, Chikusei, and Urban datasets (from different sensors with different scenarios) to verify the effectiveness of these proposed strategies in G2LHTD for HTD applications. Table I reports the stepwise performance improvement by gradually adding or changing different strategies, while Fig. 7 shows the separability map of the different models on the three datasets accordingly. First of all, to better understand the setup of the different models in the ablation experiments, we give the following description.

1) For Model_1, it just uses a dual window detector to perform target detection on the original HSI.
2) For Model_2, it first models the global spatial features of the HSI using EMAP, and then detects them using the dual window detector in Model_1.
3) For Model_3, the dual window detector in Model_2 is replaced by a D^2CEM detector to extract the eight-neighborhood information of the test pixel. Moreover, all neighborhood information is fused by arithmetic averaging, formulated as follows:

$$\boldsymbol{Y}_{\text{Ave}} = \frac{1}{9}\sum \hat{\boldsymbol{Y}}_9 \tag{22}$$

where $\boldsymbol{Y}_{\text{Ave}}$ denote detection result of model_3.
4) For Model_4, it is the G2LHTD proposed in this article, which can fully consider the relationship between the neighboring pixels and the test pixels and thus use this information more flexibly to obtain the final detection results.

It can be observed from Table I that Model_1 obtained the worst detection results in all three datasets, and combined with the detection results of Model_2, it can be concluded that EMAP can effectively boost the detection performance. Notably, in addition, the results in Fig. 7 illustrate that Model_2 obtains a better performance of target and background separation relative to Model_1, which further demonstrates that the multilevel global modeling of HSI by EMAP can indeed facilitate the separation of target and background. Furthermore, for Model_3, more neighborhood information is considered does not definitely improve the detection performance of the model, especially in the case of small detected targets (Chikusei and Urban dataset). This might be explained that more neighborhood information introduces more background information. More remarkably, however, our proposed Model_4 also exploits the eight-neighborhood information, but the ANFA fusion strategy is developed to elegantly consider the different directional neighborhood information to achieve better detection results. This also suggests that the averaging fusion approach cannot effectively utilize the neighborhood

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information in different directions. In addition, we have to point out that our proposed G2LHTD can obtain better background suppression performance when the ANFA fusion strategy is employed (see Fig. 7).

E. Algorithms Used for Comparison and Experimental Settings

To fully evaluate the detection performance of the proposed G2LHTD, eight representative baseline algorithms are selected for comparison, including ACE [18], CEM [22], hCEM [24], fractional Fourier transform (FrFT) adaptive constrained energy minimization (FACEM) [60], STD [27], CSCR [28], spectral–spatial target detection (SSTD) [48], and two-stream convolution network target detection (TSCNTD) [67]. The brief introductions and parameter configurations are detailed as follows.

1) For the ACE, it treats the target detection task as a binary hypothesis testing problem based on Gaussian distribution, then the global background estimation strategy is implemented in the image, and finally the detection is derived by generalized likelihood ratio.
2) For the CEM, the core of the algorithm is to construct a filter vector using the HSI and a prior pixel of the target to be detected, and then apply the filter vector to the detected image to obtain the detection result.
3) For the hCEM, a multilayer CEM detector is built based on the conventional CEM detector. In the implementation of this article, we set the deviation δ_k to 10^{-6}, and λ is set to 200.
4) For the FACEM, the original spectral information is mapped to the fractional domain by the Fourier transform, and then the dual window detector is applied. There are two hyperparameters that should be set which are the fractional order and the size of the dual window. Referring to the parameter settings of the original literature, the specific parameter settings for the implementation in this article are as follows. The fractional order p for the San Diego, Chikusei, Texas Coast, and Urban datasets are set to 0.10, 0.10, 0.19, and 0.19, while corresponding dual window sizes are set to (25,7), (25,3), (23,5), and (23,5).
5) For the STD, the dual window strategy is adopted to generate a background dictionary, which is then combined with a sparse representation to achieve target detection. Specifically, the optimal parameters for the dual window are set as (17, 9), (9, 7), (9, 7), and (11, 5) for aforementioned four datasets.
6) For the CSCR, sparse and cooperative representations are used to model the target and background information, respectively, and the detection results are subsequently derived from the difference between the above two. Specifically, the background dictionary is estimated using the same parameter settings as the STD algorithm. Moreover, the regularization parameters λ_1 and λ_2 are set to 10^{-1} and 10^{-2}, respectively.
7) For the SSTD, self-spectral learning with a spectral GAN is used to map the origin HSI into the deep latent space, and then a spectral-spatial framework is proposed to achieve target detection. Specifically, the proportion of selected bands ε and the degree of background suppression δ are set to 0.7 and 5.
8) For the TSCNTD, a two-stream convolution network is proposed to extract rich spectral information in HSI. Specifically, the number of selected background samples N is set to 1000 and the abundance of background material implanted in the target prior pixels is set as 0%–20%.

F. Quantitative Results and Analysis

Table II, Figs. 8 and 9 quantitatively report the detection performance of different algorithms on each of the four standard datasets in terms of AUC, ROC, and separability maps.

Specifically, the ACE and the CEM achieve similar detection results on the four datasets, with significantly lower detection performance on the Texas Coast scene than most other methods. This might be attributed to the low spatial resolution of the Texas coast dataset causing too many mixed pixels. However, since the ACE estimates the global background, it achieves a more outstanding performance in background suppression (as seen in the green box in Fig. 9). Furthermore, when multilevel CEM detectors are applied, the hCEM compensates for the poor performance of the conventional CEM detectors in low spatial resolution scenarios. Meanwhile, it can be observed from Fig. 9 that the hCEM has better background suppression compared to the CEM. Not surprisingly, FACEM improves the utilization of local spatial information by means of the four-neighborhood information of the test pixel, thus yielding high performance in terms of AUC for San Diego, Texas Coast, and Chikusei datasets. However, FACEM performs the worst detection accuracy in the Urban scene due to the fact that it assumes that the four neighborhoods of the test pixel are of the same category as the test pixel, and six of the nine targets under test in this scene consist of less than four pixels. Moreover, it can be observed from Fig. 9 that this assumption also makes the detection results on these four datasets do not separate the target and the background well. It is obvious to notice that STD obtains the worst detection results among all methods, due to the fact that performing a sparse representation of the results for dual window detection is susceptible to inducing error accumulation. We would like to point out that, moreover, the performance of the STD is considerably dependent on the detection results of the dual window detector. Unlike STD, a cooperative representation is used to model the background pixels to mitigate the errors introduced by the coarse detection of the dual window detector (especially for pixels located at the edges of the target). Therefore, the CSCR achieves better detection accuracy compared to STD. Furthermore, it can be obviously observed that the DL-based SSTD and TSCNTD do not achieve extraordinary performance (especially in the Chikusei and Urban datasets with fewer targets) due to the dependence of their performance on the training sample size. From Fig. 9, the SSTD obtains a favorable background suppression performance due to the nonlinear fusion strategy employed. However, the value of the target also suffers. Moreover, as reported in Fig. 8, the

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TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT DETECTORS IN TERMS OF AUC ON FOUR HSI DATASETS. THE BOLD ENTITIES INDICATE THE BEST ONE

Datasets	ACE	CEM	hCEM	FACEM	STD	CSCR	SSTD	TSCNTD	G2LHTD
San Diego	0.9902	0.9953	0.9905	0.9933	0.9363	0.9975	0.9868	0.9941	**0.9995**
Texas Coast	0.9501	0.9654	0.9906	0.9977	0.8711	0.9965	0.9982	0.9893	**0.9993**
Chikusei	0.9916	0.9955	0.9862	0.9945	0.9602	0.9916	0.9320	0.9441	**0.9991**
Urban	0.9824	0.9962	0.9983	0.8852	0.9786	0.9929	0.9865	0.9692	**0.9994**

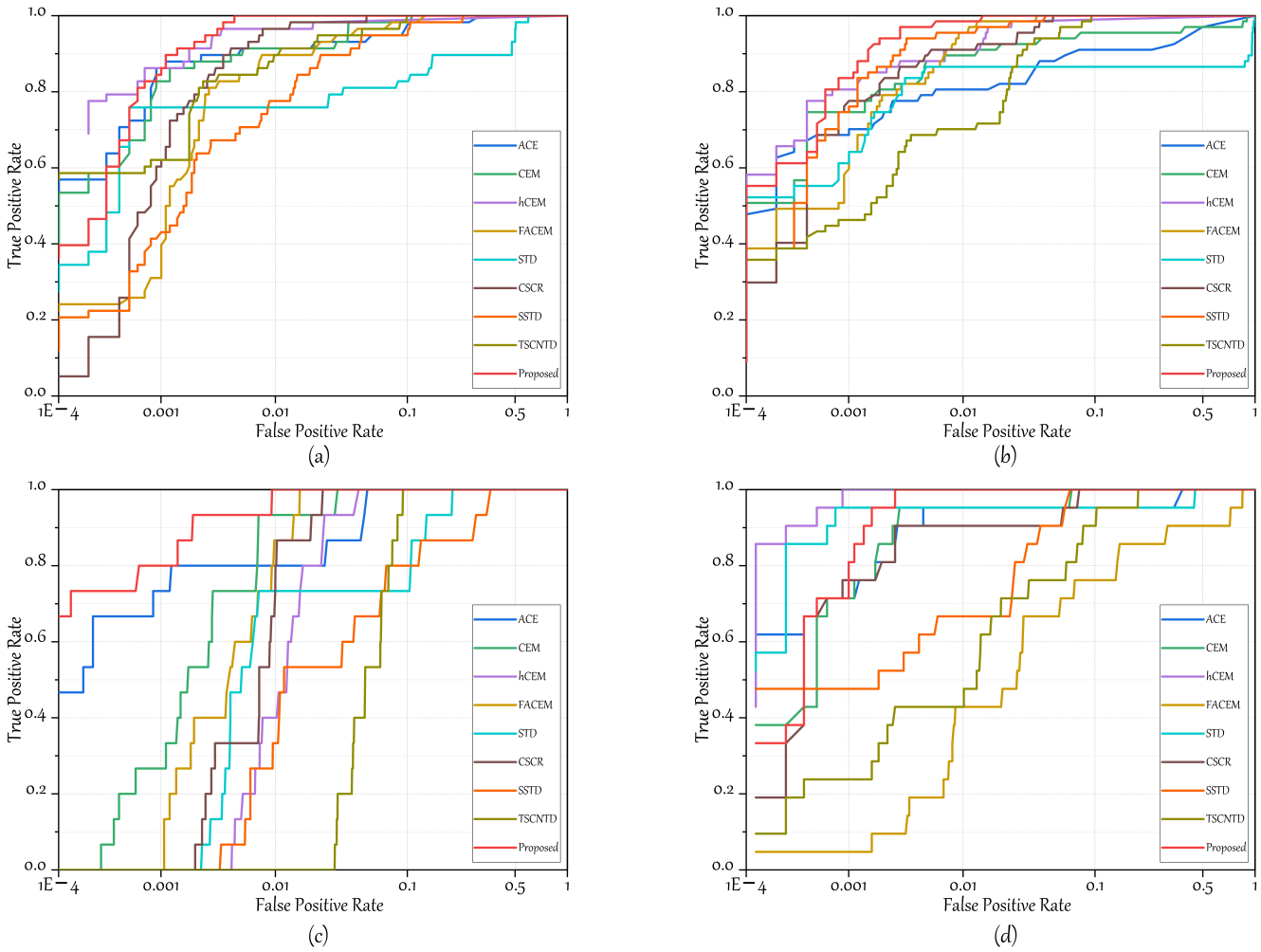

Fig. 8. Quantitative comparison of different detectors in terms of ROC on four HSI datasets. (a) San Diego. (b) Texas Coast. (c) Chikusei. (d) Urban.

TSCNTD has a high false alarm rate, which matches its poor background suppression performance shown in Fig. 9. This could perhaps be explained by a few background samples inadvertent mixed in target samples.

Remarkably, our G2LHTD obtains outstanding performance on all four datasets compared to other methods, demonstrating its superior generalization. As expected, the proposed method more flexibly handles the contribution of different directional neighborhood information to the central pixel to be tested, thus maintaining promising performance in detecting small targets in the Urban dataset. Moreover, it can be observed from Fig. 9 that the proposed method is also competitively in the separation of target and background relative to other methods, especially with similarly high levels of background suppression in all four datasets. In Fig. 8, furthermore, it can be seen that the TPR of our proposed method is almost the first to reach 1.0 on all four datasets. Although the hCEM can suppress the background better, leading to its TPR reaching 1.0 first on the Urban dataset, it can be noticed in Fig. 9 that the target value is also suppressed at a lower level by the hCEM.

G. Visual Evaluation

As illustrated in Fig. 10, we also visually compare the detection results of different detection algorithms in the form of detection maps. Overall, ACE, hCEM, and the method

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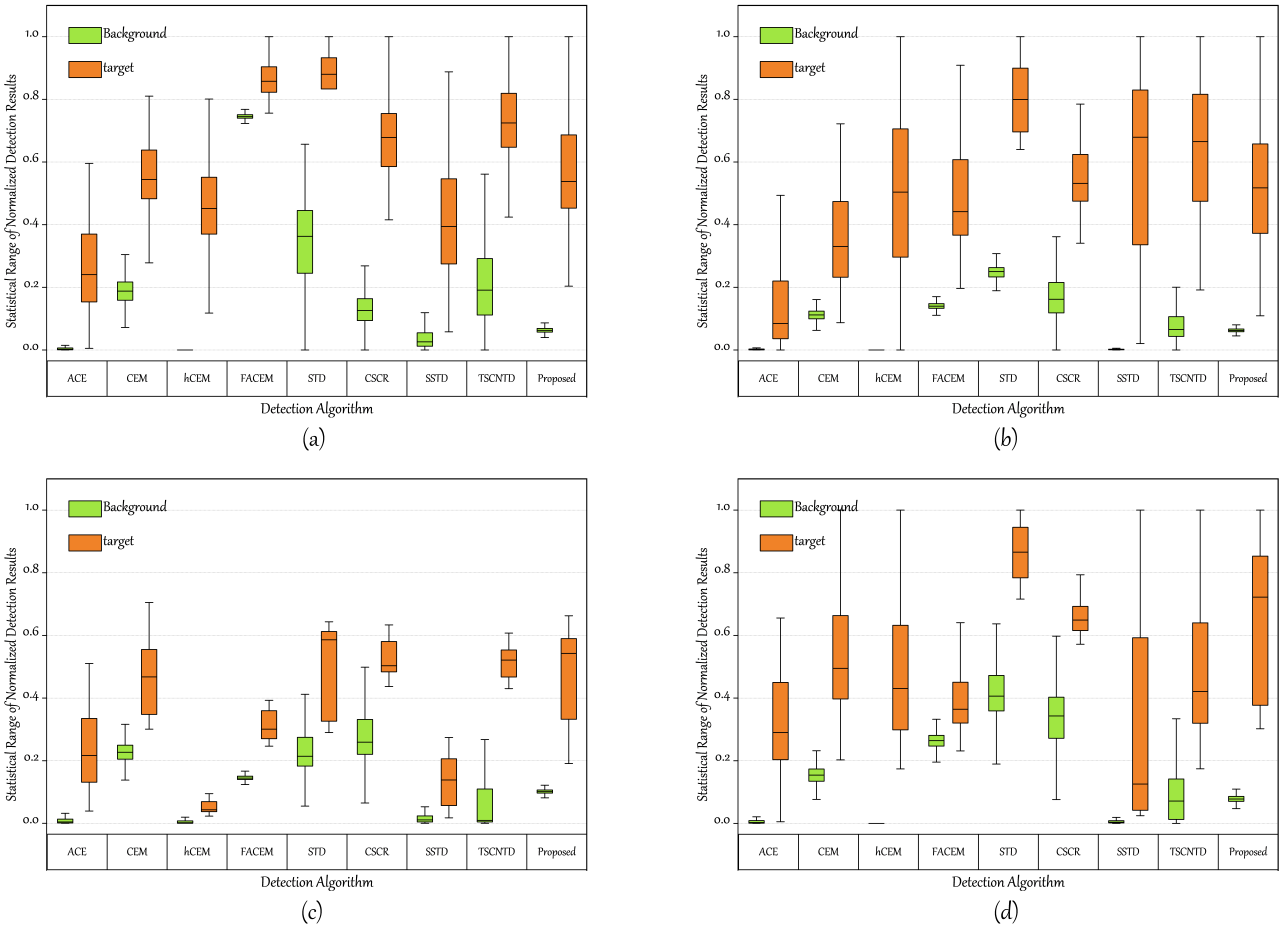

Fig. 9. Quantitative comparison of different detectors in terms of separability map on four HSI datasets. (a) San Diego. (b) Texas Coast. (c) Chikusei. (d) Urban.

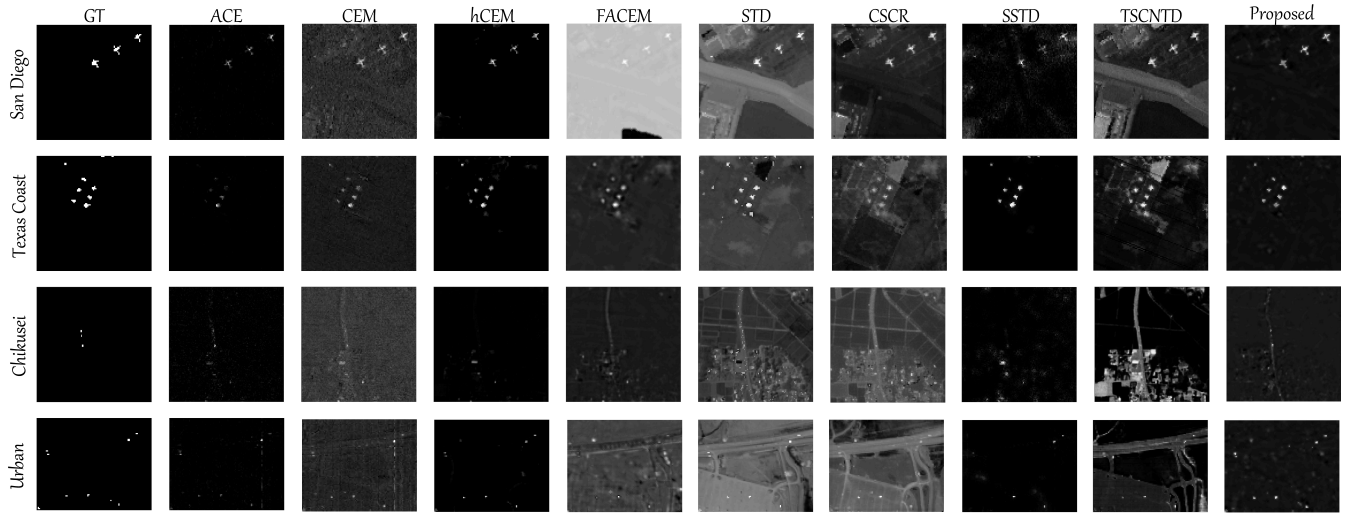

Fig. 10. Detection maps obtained by different algorithms on four HSI datasets. (a) San Diego. (b) Texas Coast. (c) Chikusei. (d) Urban.

proposed in this study have a fairly favorable level of background information suppression performance. Although the ACE, hCEM, and SSTD have slightly stronger background suppression than G2LHTD, these three methods compromise target distinguishability while obtaining better background suppression. Particularly, the ACE loses approximately half of the targets in the detection maps for the Texas Coast scene, while the hCEM and SSTD observe almost no targets in the

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detection maps for the Chikusei dataset. For CEM, STD, CSCR, and TSCNTD, although the targets are highlighted to some extent, the background noise is unacceptable and would swamp tiny targets (as observed from the Chikusei and Urban datasets). Additionally, the detection maps on all four datasets do not yield satisfactory performance due to the fact that FACEM do not handle the relationship between the neighborhood information and test pixel. EMAP can initially separate different structural information in HSI thus facilitating the separation of target and background, and ANFA strategy makes the detection at the target boundary more accurate. As expected, the proposed G2LHTD has better background suppression capability and more outstanding target detection performance in HTD.

V. Conclusion

In HTD tasks, the background complexity and the diversity of target morphology pose great challenges for improving detection performance. Accordingly, in this article, we develop a novel detection algorithm, called G2LHTD, which allows us to fully and flexibly exploit the spatial features of HSI for target detection. Specifically, EMAP is first applied to model the whole scene at various levels, thus obtaining a preliminary separation of background and target, which facilitates for further fine detection. Furthermore, we design the D^2CEM based on the original dual window detector, which achieves a comprehensive investigation of local information by considering the information of the eight neighborhoods of the test pixel. More importantly, the ANFA strategy is further proposed to flexibly exploit the neighborhood information in different directions by exploring the relationship between the neighborhood pixels, the test pixels, and the prior pixels. Our extensive experimental results, implemented on four widely used benchmarks, demonstrate that our proposed G2LHTD is effective and superior.

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